# MADHAV SINGH BHANOT

**AI Full-Stack Engineer | React, TypeScript, Node.js, Python, LLM Systems**

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## SUMMARY

Full-stack engineer (B.Tech CSE, June 2026) who builds LLM products end to end: agent UIs in React + TypeScript, real-time Node.js backends, and Airflow/Celery ingestion and LLM-eval pipelines in Python at a Helsinki AI startup; before that, a full Instagram-style social app at a Bangalore startup. Built Gaminai, an eval-driven RAG assistant with an agentic deep-research mode.

## SKILLS

- **Frontend:** React 19, TypeScript, Next.js (SSR/SSG), React Native (Expo), Vue.js, Tailwind CSS, Framer Motion, IndexedDB
- **Backend:** Node.js, Express.js, REST, WebSockets, Socket.IO, SSE, Webhooks, Pub/Sub, RBAC, OAuth 2.0, JWT, Prisma, Mongoose
- **AI & Data:** RAG, AI Agents (tool calling), pgvector Hybrid Search, LLM-as-Judge Evals, Prompt-Injection Defense, Vercel AI SDK, OpenAI/Anthropic/Gemini/Groq
- **Databases & Cloud:** PostgreSQL (migrations, RLS), Supabase, MongoDB, Docker, AWS (EC2, S3), Vercel, Render, GitHub Actions (CI)
- **Languages & Tools:** JavaScript (ES6+), TypeScript, Python, Java, SQL, Jest, Vitest, Git, Postman, Jira

## EXPERIENCE

**Innate AI** | Full-Stack / React Developer (Internship) | Helsinki, Finland (Remote) | Jul 2025 - Oct 2025

- Built the React + TypeScript interface of an LLM research platform used by VC advisory teams: step-wise agent "thinking" view, live sources panel, and Claude-style editable report artifacts on a custom Markdown editor and parser.
- Wrote the Node.js **WebSocket** layer for concurrent multi-chat rooms and a real-time **Signals** pipeline (Airflow DAGs to webhooks to pub/sub push into React), replacing HTTP polling (~300ms faster updates).
- Implemented multi-tenant **RBAC over OAuth** across API middleware and protected routes; wrote PostgreSQL migrations (Postgres and Airflow in Docker) and stored S3 media references.
- Built PDF/DOCX/HTML parsers and Airflow DAGs on autoscaling Celery workers (1,000+ jobs per worker) feeding a **RAG ingestion pipeline** (Pydantic, vector DB, JSON/Parquet).
- Built an async **LLM-as-judge eval harness** (GPT-4o-mini, Gemini 2.0 Flash) scoring retrieval and summarization, traced in LangSmith.

**Krishlabs** | Junior Full-Stack Developer (Contract) | Bangalore, India | Nov 2024 - Apr 2025

- Built an Instagram-style social app end to end (Expo React Native, Node.js/Express, MongoDB, 50+ REST endpoints): reels, posts with in-app photo editing, stories, nested comments, follow/block/privacy.
- Built a **personalized feed** ranked by interest tags, followed accounts and trending engagement, with a cold-start fallback.
- Built real-time 1:1 and group chat over **Socket.IO** (image, audio and document messages, presence) with Expo push notifications and unread counts.
- Moved media to direct-to-**S3 presigned URLs**; secured the EC2 API (Clerk + OAuth, JWT refresh, Joi, Helmet, rate limiting, XSS/NoSQL-injection sanitization); shipped the Next.js landing page.

## OPEN SOURCE

- **louislam/uptime-kuma (92k+ stars) - PR #7732, in review (2026):** Vue.js feature syncing monitor filters with URL query params so filtered views persist and are shareable; +336 lines incl. a 206-line test spec.
- **lingdojo/kana-dojo (3.6k+ stars) - PR #28654, merged (2026):** Contributed a new community UI theme, merged by maintainers.

## PROJECTS

**Gaminai - AI Research Assistant** | React, TypeScript, Express, PostgreSQL + pgvector, Vercel AI SDK | Live | Evals | GitHub

- Built a full-stack RAG research assistant streaming cited answers over SSE, with **Gemini-to-Groq model fallback**, RLS-locked Supabase data, request logging with an admin ops dashboard, and a free-tier Vercel + Render deploy with a public guest demo.
- Designed **two-stage hybrid retrieval** (Postgres full-text recall, cached Gemini embeddings, RRF fusion) with inline [n] citations: cited answers rose from 1% to 99% at 98%+ citation precision; URL-question accuracy rose from 83% to 91%.
- Shipped an **agentic deep-research mode**: an AI SDK tool-calling loop (web search, URL extraction) that decomposes multi-hop questions under step, time and token budgets, streams each step live, and cancels model calls when the user stops.
- Built an **LLM-as-judge eval harness** over a 100-question golden set plus a prompt-injection canary set (correctness, faithfulness, citation precision, false premises, p50/p95 latency), gated in CI; disliked answers become new eval cases.

**ProScape-Focus - Offline-First Focus Workspace** | React 19, IndexedDB, Express.js, OAuth 2.0 | Live | GitHub

- Built an offline-first workspace (React 19, Vite, Tailwind v4) that caches playlists and media in **IndexedDB** and syncs state through a 600ms debounced write buffer, so return visits skip the network.
- Built an Express 5 backend with Google/GitHub OAuth (Passport.js) and youtubei.js media fetching; a node-cache layer over YouTube data cut API latency by 50% and avoided rate limits.

**RunAway - Interactive Audio & Canvas Studio** | Vanilla JS, Canvas 2D, Web Audio API, Node.js | Live | GitHub

- Built a framework-free, zero-dependency animated sketchbook scene (Canvas 2D, OffscreenCanvas): procedural paper, sky and flora plus a forward-kinematics stickman rig with three gaits; 30+ daily users.
- Wrote a procedural **Web Audio** rain synthesizer (pink/brown noise) synced to on-canvas rain, and batched rain rendering to cut canvas draw calls by 98%+ at a 60/120 FPS target.
- Shipped a zero-dependency Node.js dev server, a single-file 183 KB build and a 358-test unit suite covering the PRNG, rig and scene.

## EDUCATION & ACHIEVEMENTS

**Bharati Vidyapeeth Deemed University** | Pune, India | B.Tech CSE | CGPA 8.32/10 | June 2026

- **Hackathon:** 3rd place out of 200+ onsite participants at the Dev with AI Hackathon (Unstop).